
AWS CloudTrail – Overview

What is AWS CloudTrail?

AWS CloudTrail is a **logging service** that **records all API calls** and **account activity** made within your AWS environment.

Think of it as a **"security camera" for your AWS account**, capturing *who did what, when, and from where*.

What Does It Track?

CloudTrail records activities from:

- ✓ AWS Management Console
 - ✓ AWS CLI
 - ✓ AWS SDKs and APIs
 - ✓ Other AWS services
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What Gets Logged?

Each **CloudTrail event** includes:

Field	Description
Event Time	When the action occurred
Username	Who performed the action
Source IP	Where the request came from
Event Name	What action was taken (e.g., <code>RunInstances</code>)

Resource Info What resource was impacted

How Does CloudTrail Work?

1. **Enabled by default** for management events (last 90 days).
 2. To store long-term or create custom trails:
 - Create a **Trail**
 - Send logs to **S3**
 - Optionally integrate with **CloudWatch Logs** or **SNS**
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Types of Events

Event Type	Description
Management Events	Control-plane actions (e.g., create EC2, delete bucket)
Data Events	High-volume events (e.g., S3 GetObject , Lambda Invoke)
Insight Events	Detect unusual activity (requires CloudTrail Insights)

Common Use Cases

-  **Security Auditing**
Detect unauthorized access or changes
-  **Compliance and Governance**
Meet regulations like PCI-DSS, HIPAA, SOC 2

-  **Operational Troubleshooting**
Trace changes to resources during incidents
-  **Behavior Analysis**
Who deleted an EC2 instance? When was a bucket made public?

Pricing (as of latest AWS guidelines)

Feature	Cost
Event History (90d)	Free
Custom Trails	\$2 per 100,000 events
Data Events	Charged separately
CloudTrail Insights	Extra charge per event

Example: View Event History

From Console:

AWS CloudTrail → **Event History**

From CLI:

```
aws cloudtrail lookup-events \  
--max-results 5 \  
--query 'Events[*].{Time:EventTime,Action:EventName,User:Username}'
```

Best Practices

- ✓ Enable CloudTrail in **all regions**
 - ✓ Store logs in **S3 with encryption**
 - ✓ Enable **multi-trail setup** (e.g., org-wide trail)
 - ✓ Send events to **CloudWatch Logs for alerting**
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Summary

Feature	Description
What	Monitors all AWS API activity
How	Logs stored in S3, viewable via console or CLI
Why	Audit, troubleshoot, secure
